

G3 Applications of differentiation

Name: _____

Class: _____

Date: _____

Time: **420 minutes**

Marks: **349 marks**

Comments:

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Q1.

A curve has equation $3xy - 2y^2 = 4$.

Find the gradient of the curve at the point (2, 1).

(Total 5 marks)

Q2.

Two numbers, x and y , are such that $3x + y = 9$, where $x \geq 0$ and $y \geq 0$.

It is given that $V = xy^2$.

(a) Show that $V = 81x - 54x^2 + 9x^3$.

(2)

(b) (i) Show that $\frac{dV}{dx} = k(x^2 - 4x + 3)$, and state the value of the integer k .

(4)

(ii) Hence find the two values of x for which $\frac{dV}{dx} = 0$.

(2)

(c) Find $\frac{d^2V}{dx^2}$.

(2)

(d) (i) Find the value of $\frac{d^2V}{dx^2}$ for each of the two values of x found in part (b)(ii).

(1)

(ii) Hence determine the value of x for which V has a maximum value.

(1)

(iii) Find the maximum value of V .

(1)

(Total 13 marks)

Q3.

(a) Write $\sqrt{x^3}$ in the form x^k , where k is a fraction.

(1)

(b) A curve, defined for $x \geq 0$, has equation

$$y = x^2 - \sqrt{x^3}$$

(i) Find $\frac{dy}{dx}$.

(3)

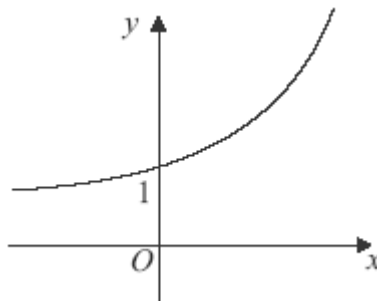
- (ii) Find the equation of the tangent to the curve at the point where $x = 4$, giving your answer in the form $y = mx + c$.

(5)

(Total 9 marks)

Q4.

The diagram shows a sketch of the curve with equation $y = 6^x$.



- (a) (i) Use the trapezium rule with five ordinates (four strips) to find an approximate value for $\int_0^2 6^x dx$, giving your answer to three significant figures.

(4)

- (ii) Explain, with the aid of a diagram, whether your approximate value will be an overestimate or an underestimate of the true value of $\int_0^2 6^x dx$.

(2)

- (b) (i) Describe a single geometrical transformation that maps the graph of $y = 6^x$ onto the graph of $y = 6^{3x}$.

(2)

- (ii) The line $y = 84$ intersects the curve $y = 6^{3x}$ at the point A . By using logarithms, find the x -coordinate of A , giving your answer to three decimal places.

(4)

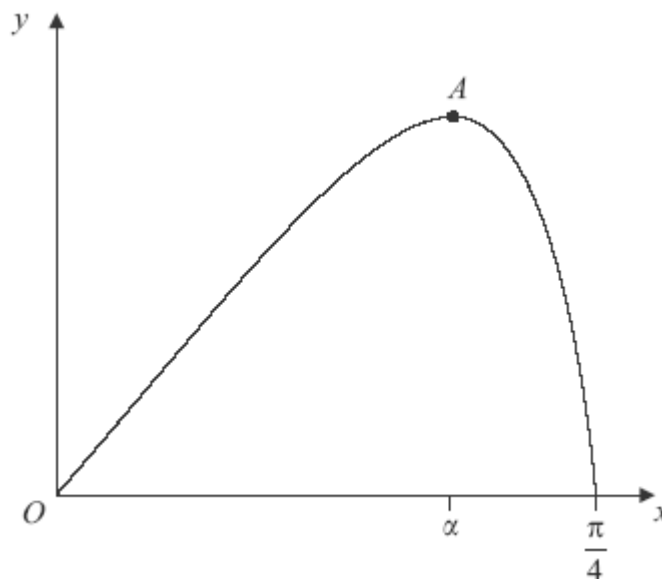
- (c) The graph of $y = 6^x$ is translated by $\begin{bmatrix} 1 \\ -2 \end{bmatrix}$ to give the graph of the curve with equation $y = f(x)$. Write down an expression for $f(x)$.

(2)

(Total 14 marks)

Q5.

A curve is defined for $0 \leq x \leq \frac{\pi}{4}$ by the equation $y = x \cos 2x$, and is sketched below.



- (a) Find $\frac{dy}{dx}$. (2)
- (b) The point A , where $x = \alpha$, on the curve is a stationary point.
- (i) Show that $1 - 2\alpha \tan 2\alpha = 0$. (2)
- (ii) Show that $0.4 < \alpha < 0.5$. (2)
- (iii) Show that the equation $1 - 2x \tan 2x = 0$ can be rearranged to become $x = \frac{1}{2} \tan^{-1} \left(\frac{1}{2x} \right)$. (1)
- (iv) Use the iteration $x_{n+1} = \frac{1}{2} \tan^{-1} \left(\frac{1}{2x_n} \right)$ with $x_1 = 0.4$ to find x_3 , giving your answer to two significant figures. (2)
- (c) Use integration by parts to find $\int_0^{0.5} x \cos 2x dx$, giving your answer to three significant figures. (5)

(Total 14 marks)

Q6.

A curve is defined, for $t \neq 0$, by the parametric equations

$$x = 4t + 3, \quad y = \frac{1}{2t} - 1$$

At the point P on the curve, $t = \frac{1}{2}$.

(a) Find the gradient of the curve at the point P .

(4)

(b) Find an equation of the normal to the curve at the point P .

(3)

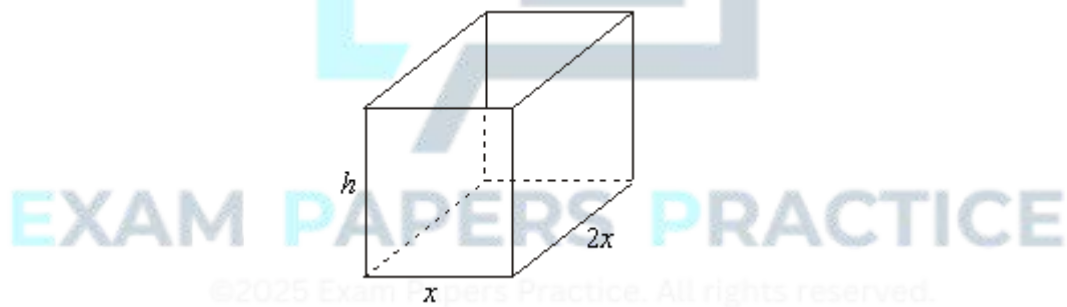
(c) Find a cartesian equation of the curve.

(3)

(Total 10 marks)

Q7.

The diagram shows an **open-topped** water tank with a horizontal rectangular base and four vertical faces. The base has width x metres and length $2x$ metres, and the height of the tank is h metres.



The combined internal surface area of the base and four vertical faces is 54 m^2 .

(a) (i) Show that $x^2 + 3xh = 27$.

(2)

(ii) Hence express h in terms of x .

(1)

(iii) Hence show that the volume of water, $V \text{ m}^3$, that the tank can hold when full is given by

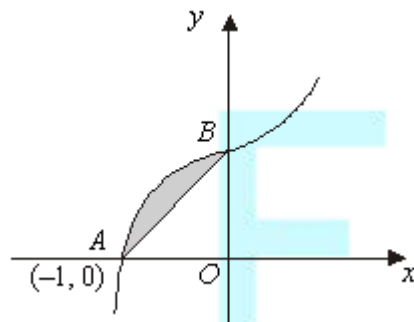
$$V = 18x - \frac{2x^3}{3}$$

(1)

- (b) (i) Find $\frac{dV}{dx}$. (2)
- (ii) Verify that V has a stationary value when $x = 3$. (2)
- (c) Find $\frac{d^2V}{dx^2}$ and hence determine whether V has a maximum value or a minimum value when $x = 3$. (2)
- (Total 10 marks)

Q8.

The curve with equation $y = 3x^5 + 2x + 5$ is sketched below.

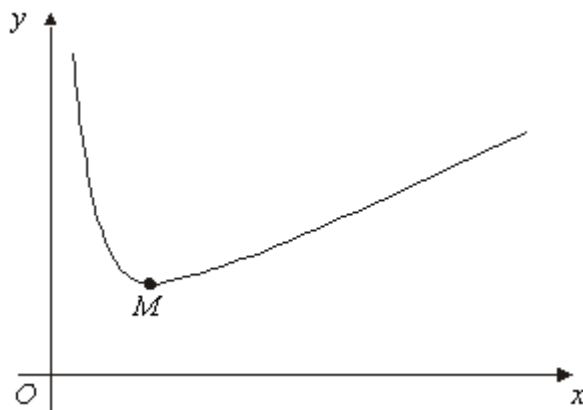


The curve cuts the x -axis at the point $A (-1, 0)$ and cuts the y -axis at the point B .

- (a) (i) State the coordinates of the point B and hence find the area of the triangle AOB , where O is the origin. (3)
- (ii) Find $\int (3x^5 + 2x + 5) dx$ (3)
- (iii) Hence find the area of the shaded region bounded by the curve and the line AB . (4)
- (b) (i) Find the gradient of the curve with equation $y = 3x^5 + 2x + 5$ at the point $A(-1, 0)$. (3)
- (ii) Hence find an equation of the tangent to the curve at the point A . (1)
- (Total 14 marks)

Q9.

A curve C is defined for $x > 0$ by the equation $y = x + 1 + \frac{4}{x^2}$ and is sketched below.



- (a) (i) Given that $y = x + 1 + \frac{4}{x^2}$, find $\frac{dy}{dx}$. (3)
- (ii) The curve C has a minimum point M . Find the coordinates of M . (4)
- (iii) Find an equation of the normal to C at the point $(1, 6)$. (4)
- (b) (i) Find $\int \left(x + 1 + \frac{4}{x^2} \right) dx$. (3)
- (ii) Hence find the area of the region bounded by the curve C , the lines $x = 1$ and $x = 4$ and the x -axis. (2)
- (Total 16 marks)**

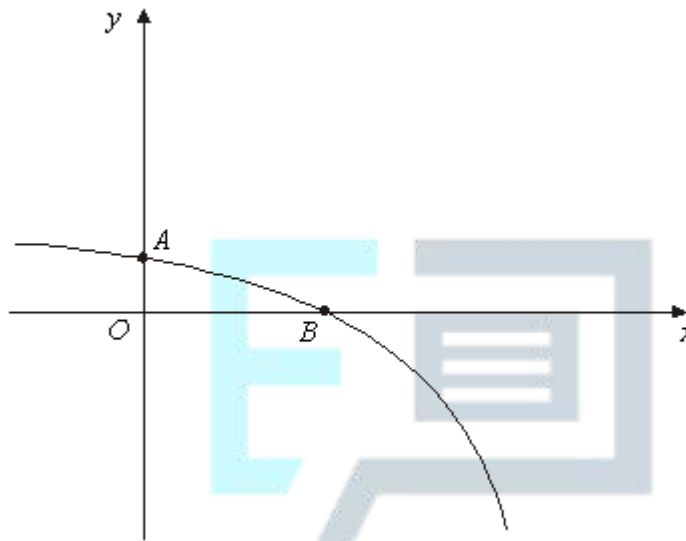
Q10.

- (a) Find $\frac{dy}{dx}$ when:
- (i) $y = (4x^2 + 3x + 2)^{10}$; (2)
- (ii) $y = x^2 \tan x$. (2)

- (b) (i) Find $\frac{dx}{dy}$ when $x = 2y^3 + \ln y$. (1)
- (ii) Hence find an equation of the tangent to the curve $x = 2y^3 + \ln y$ at the point (2,1). (3)
- (Total 8 marks)

Q11.

The sketch shows the graph of $y = 4 - e^{2x}$. The curve crosses the y -axis at the point A and the x -axis at the point B .



- (a) (i) Find $\int (4 - e^{2x}) dx$. (2)
- (ii) Hence show that $\int_0^{\ln 2} (4 - e^{2x}) dx = 4 \ln 2 - \frac{3}{2}$. (2)
- (b) (i) Write down the y -coordinate of A . (1)
- (ii) Show that $x = \ln 2$ at B . (2)
- (c) Find the equation of the normal to the curve $y = 4 - e^{2x}$ at the point B . (4)
- (d) Find the area of the region enclosed by the curve $y = 4 - e^{2x}$, the normal to the curve at B and the y -axis.

(3)
 (Total 14 marks)

Q12.

A curve is defined by the parametric equations

$$x = 1 + 2t, \quad y = 1 - 4t^2$$

(a) (i) Find $\frac{dx}{dt}$ and $\frac{dy}{dt}$.

(2)

(ii) Hence find $\frac{dy}{dx}$ in terms of t .

(2)

(b) Find an equation of the normal to the curve at the point where $t = 1$.

(4)

(c) Find a cartesian equation of the curve.

(3)

(Total 11 marks)

Q13.

A model helicopter takes off from a point O at time $t = 0$ and moves vertically so that its height, y cm, above O after time t seconds is given by

$$y = \frac{1}{4}t^4 - 26t^2 + 96t, \quad 0 \leq t \leq 4$$

(a) Find:

(i) $\frac{dy}{dt}$;

(3)

(ii) $\frac{d^2y}{dt^2}$

(2)

(b) Verify that y has a stationary value when $t = 2$ and determine whether this stationary value is a maximum value or a minimum value.

(4)

(c) Find the rate of change of y with respect to t when $t = 1$

(2)

- (d) Determine whether the height of the helicopter above O is increasing or decreasing at the instant when $t = 3$.

(2)

(Total 13 marks)

Q14.

A curve is defined for $x > 0$ by the equation

$$y = \left(1 + \frac{2}{x}\right)^2$$

The point P lies on the curve where $x = 2$.

- (a) Find the y -coordinate of P .

(1)

- (b) Expand $\left(1 + \frac{2}{x}\right)^2$.

(2)

- (c) Find $\frac{dy}{dx}$

(3)

- (d) Hence show that the gradient of the curve at P is -2 .

(2)

- (e) Find the equation of the normal to the curve at P , giving your answer in the form $x + by + c = 0$, where b and c are integers.

(4)

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(Total 12 marks)

Q15.

- (a) Differentiate $\ln x$ with respect to x .

(1)

- (b) Given that $y = (x + 1) \ln x$, find $\frac{dy}{dx}$.

(2)

- (c) Find an equation of the normal to the curve $y = (x + 1) \ln x$ at the point where $x = 1$.

(4)

(Total 7 marks)

Q16.

(a) A curve has equation $y = (x^2 - 3)e^x$.

(i) Find $\frac{dy}{dx}$.

(2)

(ii) Find $\frac{d^2y}{dx^2}$.

(2)

(b) (i) Find the x -coordinate of each of the stationary points of the curve.

(4)

(ii) Using your answer to part (a)(ii), determine the nature of each of the stationary points.

(2)

(Total 10 marks)

Q17.

The point $P(1, a)$, where $a > 0$, lies on the curve $y + 4x = 5x^2y^2$.

(a) Show that $a = 1$.

(2)

(b) Find the gradient of the curve at P .

(7)

(c) Find an equation of the tangent to the curve at P .

(1)

(Total 10 marks)

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Q18.

The volume, $V \text{ m}^3$, of water in a tank at time t seconds is given by

$$V = \frac{1}{3} t^6 - 2t^4 + 3t^2, \quad \text{for } t \geq 0$$

(a) Find:

(i) $\frac{dV}{dt}$;

(3)

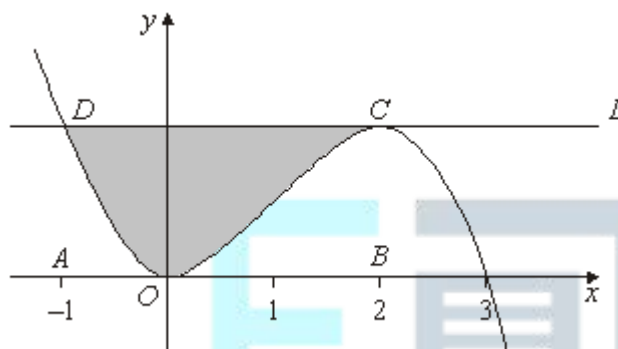
(ii) $\frac{d^2V}{dt^2}$.

- (2)
- (b) Find the rate of change of the volume of water in the tank, in $\text{m}^3 \text{s}^{-1}$, when $t = 2$. (2)
- (c) (i) Verify that V has a stationary value when $t = 1$. (2)
- (ii) Determine whether this is a maximum or minimum value. (2)

(Total 11 marks)

Q19.

The diagram shows the curve with equation $y = 3x^2 - x^3$ and the line L .



The points A and B have coordinates $(-1, 0)$ and $(2, 0)$ respectively. The curve touches the x -axis at the origin O and crosses the x -axis at the point $(3, 0)$. The line L cuts the curve at the point D where $x = -1$ and touches the curve at C where $x = 2$.

- (a) Find the area of the rectangle $ABCD$. (2)
- (b) (i) Find $\int (3x^2 - x^3) dx$. (3)
- (ii) Hence find the area of the shaded region bounded by the curve and the line L . (4)
- (c) For the curve above with equation $y = 3x^2 - x^3$:
- (i) find $\frac{dy}{dx}$; (2)
- (ii) hence find an equation of the tangent at the point on the curve where $x = 1$; (3)

(iii) show that y is decreasing when $x^2 - 2x > 0$.

(2)

(d) Solve the inequality $x^2 - 2x > 0$.

(2)

(Total 18 marks)

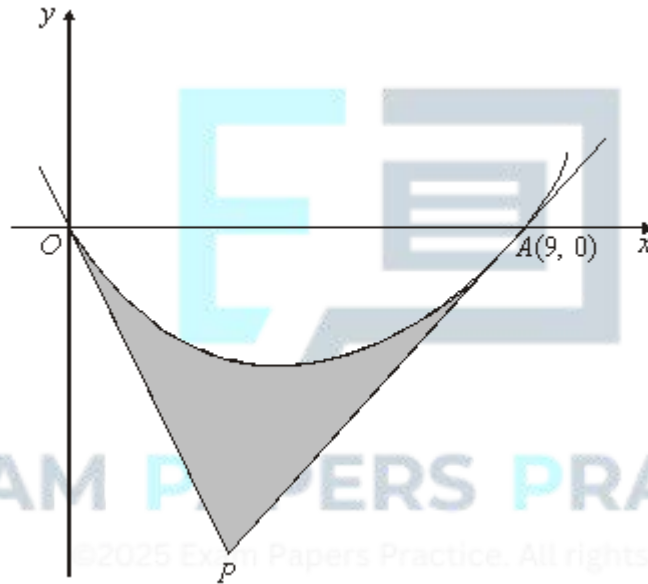
Q20.

Given that $y = 16x + x^{-1}$, find the two values of x for which $\frac{dy}{dx} = 0$.

(Total 5 marks)

Q21.

A curve, drawn from the origin O , crosses the x -axis at the point $A(9, 0)$. Tangents to the curve at O and A meet at the point P , as shown in the diagram.



The curve, defined for $x \geq 0$, has equation

$$y = x^{\frac{3}{2}} - 3x$$

(a) Find $\frac{dy}{dx}$.

(2)

(b) (i) Find the value of $\frac{dy}{dx}$ at the point O and hence write down an equation of the tangent at O .

(2)

(ii) Show that the equation of the tangent at $A(9, 0)$ is $2y = 3x - 27$.

(3)

(iii) Hence find the coordinates of the point P where the two tangents meet.

(3)

(c) Find $\int \left(x^{\frac{3}{2}} - 3x \right) dx$.

(3)

(d) Calculate the area of the shaded region bounded by the curve and the tangents OP and AP .

(5)

(Total 18 marks)

Q22.

A curve is defined by the parametric equations

$$x = 3 - 4t \quad y = 1 + \frac{2}{t}$$

(a) Find $\frac{dy}{dx}$ in terms of t .

(4)

(b) Find the equation of the tangent to the curve at the point where $t = 2$, giving your answer in the form $ax + by + c = 0$, where a , b and c are integers.

(4)

(c) Verify that the cartesian equation of the curve can be written as

$$(x - 3)(y - 1) + 8 = 0$$

(3)

(Total 11 marks)

Q23.

A curve has equation $y = 7 - 2x^5$.

(a) Find $\frac{dy}{dx}$.

(2)

(b) Find an equation for the tangent to the curve at the point where $x = 1$.

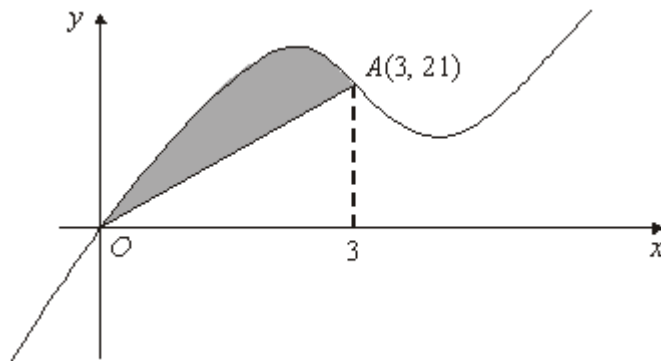
(3)

(c) Determine whether y is increasing or decreasing when $x = -2$.

(2)
(Total 7 marks)

Q24.

The curve with equation $y = x^3 - 10x^2 + 28x$ is sketched below.



The curve crosses the x -axis at the origin O and the point $A(3, 21)$ lies on the curve.

- (a) (i) Find $\frac{dy}{dx}$. (3)

- (ii) Hence show that the curve has a stationary point when $x = 2$ and find the x -coordinate of the other stationary point. (4)

- (b) (i) Find $\int (x^3 - 10x^2 + 28x) dx$. (3)

- (ii) Hence show that $\int_0^3 (x^3 - 10x^2 + 28x) dx = 56\frac{1}{4}$. (2)

- (iii) Hence determine the area of the shaded region bounded by the curve and the line OA . (3)
- (Total 15 marks)

Q25.

At the point (x, y) , where $x > 0$, the gradient of a curve is given by

$$\frac{dy}{dx} = 3x^{\frac{1}{2}} + \frac{16}{x^2} - 7$$

- (a) (i) Verify that $\frac{dy}{dx} = 0$ when $x = 4$. (1)
- (ii) Write $\frac{16}{x^2}$ in the form $16x^k$, where k is an integer. (1)
- (iii) Find $\frac{d^2y}{dx^2}$. (3)
- (iv) Hence determine whether the point where $x = 4$ is a maximum or a minimum, giving a reason for your answer. (2)
- (b) The point $P(1, 8)$ lies on the curve.
- (i) Show that the gradient of the curve at the point P is 12. (1)
- (ii) Find an equation of the normal to the curve at P . (3)
- (c) (i) Find $\int (3x^{\frac{1}{2}} + \frac{16}{x^2} - 7) dx$ (3)
- (ii) Hence find the equation of the curve which passes through the point $P(1, 8)$. (3)
- (Total 17 marks)**

Q26.

A curve is defined by the equation

$$y^2 - xy + 3x^2 - 5 = 0$$

- (a) Find the y -coordinates of the two points on the curve where $x = 1$. (3)
- (b) (i) Show that $\frac{dy}{dx} = \frac{y - 6x}{2y - x}$. (6)
- (ii) Find the gradient of the curve at each of the points where $x = 1$. (2)

- (iii) Show that, at the two stationary points on the curve, $33x^2 - 5 = 0$.

(3)

(Total 14 marks)

Q27.

A curve has equation $y = x^5 - 6x^3 - 3x + 25$.

- (a) Find $\frac{dy}{dx}$.

(3)

- (b) The point P on the curve has coordinates $(2, 3)$.

- (i) Show that the gradient of the curve at P is 5.

(2)

- (ii) Hence find an equation of the normal to the curve at P , expressing your answer in the form $ax + by = c$, where a , b and c are integers.

(3)

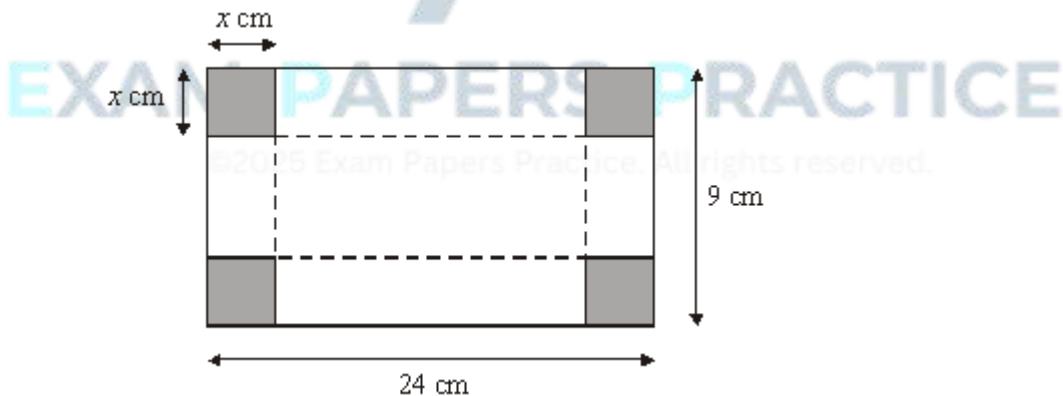
- (c) Determine whether y is increasing or decreasing when $x = 1$.

(2)

(Total 10 marks)

Q28.

The diagram below shows a rectangular sheet of metal 24 cm by 9 cm.



A square of side x cm is cut from each corner and the metal is then folded along the broken lines to make an open box with a rectangular base and height x cm.

- (a) Show that the volume, V cm³, of liquid the box can hold is given by

$$V = 4x^3 - 66x^2 + 216x$$

(3)

- (b) (i) Find $\frac{dV}{dx}$. (3)
- (ii) Show that any stationary values of V must occur when $x^2 - 11x + 18 = 0$ (2)
- (iii) Solve the equation $x^2 - 11x + 18 = 0$. (2)
- (iv) Explain why there is only one value of x for which V is stationary. (1)
- (c) (i) Find $\frac{d^2V}{dx^2}$. (2)
- (ii) Hence determine whether the stationary value is a maximum or minimum. (2)
- (Total 15 marks)**

Q29.

A curve is defined for $x > 0$ by the equation $y = x + \frac{2}{x}$.

- (a) (i) Find $\frac{dy}{dx}$. (3)
- (ii) Hence show that the gradient of the curve at the point P where $x = 2$ is $\frac{1}{2}$. (1)
- (b) Find an equation of the normal to the curve at this point P . (4)
- (Total 8 marks)**

Q30.

- (a) (i) Using the binomial expansion, or otherwise, express $(2 + x)^3$ in the form $8 + ax + bx^2 + x^3$, where a and b are integers. (3)
- (ii) Write down the expansion of $(2 - x)^3$. (2)
- (b) Hence show that $(2 + x)^3 - (2 - x)^3 = 24x + 2x^3$.

(2)

- (c) Hence show that the curve with equation

$$y = (2 + x)^3 - (2 - x)^3$$

has no stationary points.

(3)

(Total 10 marks)



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